

Final Data Cleaning & Quality Report

Audit of Data Pipeline Output: notebooks/02_data_cleaning.ipynb

Evidence Source: Transcribed strictly from saved code, Markdown, and cell outputs. No cells were executed for this report. Missing or unavailable values are explicitly flagged as *Non disponible dans les résultats existants*.

1. EXECUTIVE SUMMARY

The audit documents an initial dataset of **2,294 rows and 23 columns**. Following standard cleaning protocols, one exact duplicate row and one record lacking an internship title were removed, establishing a final working shape of **2,292 rows and 23 columns**.

2,292 FINAL ROWS	23 TOTAL FIELDS	1,419 UNIQUE STUDENTS	1,105 CLEANED COMPANIES
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Key pipeline treatments include manual overrides for company postal codes and internship locations, algorithmic normalization and fuzzy matching on corporate entity names, and rigorous diagnostics covering student identifiers, agreement conventions, and dates. Substantial missingness persists in secondary location fields. Company merging relies on a hybrid of automated similarity grouping and explicit manual overrides, with specific exclusions enforced.

Important Reproducibility Limitation: The notebook contains execution traces from non-linear runs and an out-of-sequence reload of exported workbooks. The saved outputs do not verify that all in-memory transformations were fully persisted to `data/processed/stages_cleaned.xlsx`.

2. DATASET OVERVIEW AND BASE ROW OPERATIONS

The primary dataset was loaded into `df_raw` and copied to `df`. Standard filtering resulted in the removal of 2 rows total.

CHECK / AUDIT	BEFORE	AFTER	CHANGE DETAILS & OPERATIONAL NOTES
Rows & Columns	2,294 × 23	2,292 × 23	Exact duplicate and missing title record purged.
Exact Duplicate Rows	1	0	Programmatic deletion via <code>drop_duplicates()</code> .
Missing Stage Title	1	0	Filtered out using <code>dropna(subset=['Titre du stage'])</code> .
Final Export Shape	—	2,292 × 23	Explicitly reported at final export cell execution.

Categorical Distributions Recorded in Notebook

Stage Types: PFE (963), SIB (562), SIA (530), MASTER (73), SNO (60), SD (51), PFE_MASTER (46), PFE_CPRO (7).
Academic Years: 2020/2021 (360), 2021/2022 (369), 2022/2023 (375), 2023/2024 (379), 2024/2025 (350), 2025/2026 (355), 2026/2027 (100), Missing (4).

3. CHRONOLOGICAL CLEANING & MANUAL OVERRIDES

3.1 & 3.2 Programmatic Record Deduplication

Initial cleaning applied explicit rules to purge non-viable rows. One exact duplicate was removed, followed by the deletion of row index displaying a null `Titre du stage`, leaving 2,292 valid records.

3.3 Manual Postal Code Corrections

Explicit scalar assignments were applied to `Code postal`. Prior values were not captured in the saved execution state.

ROW INDEX	ASSIGNED CODE POSTAL	REASONING / CONTEXT DOCUMENTED
1	EX5 2FN	Derived from address string: 6 BABBAGE WAY, EXETER, EX5 2FN.
467	BP 0000	Non-classical postal reference; reliable code unavailable.
844	32901	Manual correction; no explicit reason logged in notebook.
848	SN15 1GD	Manual correction; no explicit reason logged in notebook.
1124	76131	Assigned alongside location corrections.
1591	H1V 3N7	Manual correction; no explicit reason logged in notebook.

3.4 & 3.5 Internship Location Adjustments (Country & Postal Code)

Direct updates to `Pays (Lieu Stage)` and `CP (Lieu Stage)` were applied to specific row indices. Several postal assignments represent regional approximations and require qualification.

INDEX	NEW COUNTRY	NEW POSTAL CODE	NOTEBOOK ANNOTATION / NOTES
6, 313	France	—	Country set to France; prior state unrecorded.
467	—	BP 0000	Lomé, Togo; non-standard postal code structure.
839	—	330-0853	Saitama, Japan; marked as approximate .
1166, 1272	France	86000 / 33700	Poitiers / Mérignac, France.
1383, 2000	—	1400 / 2926	Yverdon-les-Bains / Boncourt, Switzerland.
1496	—	050000	Hebei, China; regional approximation .
1673, 1684	Suède	752 00	Uppsala, Sweden; marked as approximate .
1763, 1976	—	434000	Jingzhou, China.
1784, 1835	Allemagne / —	— / 66000	Catalonia, Spain code marked approximate .

3.6 Multi-field Location & Company Overrides

INDEX	TARGET FIELD	ASSIGNED VALUE	TREATMENT LOG
726	Ville / Pays / Entreprise	Arc-lès-Gray / FRANCE / SIMU S.A.S	Overwrote earlier assignment ('Gray').
1124	Ville / Pays / CP	Karlsruhe / ALLEMAGNE / 76131	Filled completely missing location block.
1272	Entreprise (Lieu	THAI ES DMS FRANCE CAMPUS	Resolved missing secondary company

4. MISSING-VALUE ANALYSIS

The table below reflects the saved state of missingness in the notebook. This baseline was generated prior to certain manual location updates; complete post-treatment missingness totals are *Non disponible dans les résultats existants*.

FIELD NAME	MISSING COUNT	MISSING %	NOTEBOOK STATUS / TREATMENT
Entreprise (Lieu Stage)	1,747	76.22%	Selected manual fills; large gap remains.
CP (Lieu Stage)	1,744	76.09%	Selected manual fills; large gap remains.
Ville (Lieu Stage)	1,744	76.09%	Selected manual fills; large gap remains.
Pays (Lieu Stage)	1,646	71.82%	Selected manual fills; large gap remains.
Date soutenance	704	30.72%	Investigated; no imputation performed.
Nature du stage	453	19.76%	Investigated; left un-imputed.
Entreprise	32	1.40%	Investigated; preserved as missing.
Référent pédagogique (mail) [.1]	13	0.57%	Both referee email columns identical.
Descriptif du Stage	12	0.52%	No text content logged.
Date fin	12	0.52%	Date checks run; no fill.
Date début	11	0.48%	Date checks run; no fill.
Code postal	5	0.22%	Manual corrections applied to 5 rows.
Année académique	4	0.17%	Unresolved gaps.
Adresse	1	0.04%	Single missing record.
Complete Fields (0 missing)	0	0.00%	Type, Durée, Convention, N° étudiant, Names, Ville.

5. INTEGRITY CHECKS: IDENTITY & CONVENTIONS

- **Student Identity Mapping:** Post-export analysis identified **1,419 unique student numbers** across 2,292 rows.
- **Multiple Internships:** 778 records represent repeated student entries. This was validated as legitimate (students undertaking multiple sequential stages) and retained.
- **Name Consistency:** Grouped verification of student numbers against first/last names returned **0 identity inconsistencies**.
- **Convention Uniqueness:** Cross-referencing conventions against student IDs returned **0 multi-student convention overlaps**.

6. CHRONOLOGICAL INTEGRITY & DURATION AUDITS

- **Logical Inversion:** Evaluation of Date début > Date fin returned **0 erroneous instances**.
- **Duration Variance:** A derived metric duree_calculée_mois (date delta in days / 30.44) was compared against reported Durée totale (mois). Cases exceeding a 0.15-month variance were flagged as diagnostic findings but were not programmatically altered.
- **Outlier Records:** Zero-duration records were isolated; exact counts of negative or >24 month durations are *Non disponible dans les résultats existants*.

7. COMPANY NAME STANDARDISATION & ENTITY RESOLUTION

7.1 Normalization Pipeline

The dataset initially contained **1,199 unique non-null company names**. Standard programmatic transformations reduced this to **1,183 unique entities**:

1. Trim uppercase conversion.
2. Unicode decomposition (accent mark removal).
3. Standardization of typographic apostrophes.
4. Legal entity normalization (e.g., S . A . → SA).
5. Hyphen replacement with whitespace.
6. Whitespace collapsing and final trimming.

7.2 Fuzzy Clustering Thresholds (RapidFuzz Ratio)

SIMILARITY BAND	RESULTING UNIQUE ENTITIES	EXECUTION & GROUPING DETAILS
≥ 95%	1,165 companies	Union-find clustering. Pair count unrecorded.
90% to < 95%	1,137 companies	Union-find clustering with 2 explicit exclusions.
85% to < 90%	1,116 companies	44 candidate pairs listed; manual dictionary applied.
80% to < 85%	1,106 companies	Manual replacement dictionary applied.
76% to < 77%	1,105 companies	Targeted manual dictionary applied.

7.3 Explicit Manual Company Mappings

85% – 90% Threshold Mappings

ORIGINAL VALUE	TARGET STANDARD
THE UNIV. OF ELECTRO COMM...	THE UNIV. OF ELECTRO COMM.
FIT (FLORIDA INSITUTE OF TECH)	FLORIDA INSTITUTE OF TECH.
FLORIDA INSTITUTE OF TECH	FLORIDA INSTITUTE OF TECH.
OFF NAT ETUDES RECH AERO...	OFF NAT ETUDES RECH AEROSPATIALE
INSTITUT NAT RECHERCHE SECURITE	INSTITUT NAT DE RECHERCHE...
CARAN D'ACHE	CARAN D'ACHE SA
KERI MEDICAL	KERI MEDICAL SA
ALSTOM TRANSPORT SA (93)	ALSTOM TRANSPORT SA
HERMES SELLIER SAS	HERMES SELLIER
GE HEALTHCARE SAS	GE HEALTHCARE
DECATHLON SE	DECATHLON

80% – 85% & 76% – 77% Mappings

ORIGINAL VALUE	TARGET STANDARD
KOMATSUSEIKI KOSAKUSHO CO...	KOMATSUSEIKI KOSAKUSHO
THALES ALENIA SPACE FRANCE	THALES ALENIA SPACE
BRUNEL UNIVERSITY LONDON	BRUNEL UNIVERSITY
MANUFACTURE HORLOGERE VAL...	MANUFACTURE VALFLEURIER
ARIANEGROUP SAS	ARIANE GROUP
BAUD INDUSTRIES SUISSE	BAUD INDUSTRIES
R.MONTAVON SA	REMY MONTAVON SA.
PARALEC ENERGY CO LTD	PARALEC ENERGY
UNIVERSIDAD DE OVIEDO	UNIVERSITY OF OVIEDO
MANUFACTURE HORLOGERE...	MANUFACTURE HORLOGERE...

7.4 Explicitly Excluded Merges

7.5 Comparison of Primary & Secondary Company Fields

An exploratory comparison was executed on **537 records** possessing non-null values for both Entreprise and Entreprise (Lieu Stage):

- **Mean Similarity Score:** 59.59% | **Median Similarity Score:** 59.09%
- **Distribution:** 203 pairs ≥ 90% similarity; 224 pairs ≥ 80%; 249 pairs ≥ 70%.

Diagnostic Conclusion: High variance confirms these columns represent distinct concepts (legal contracting entity vs. physical operational site) and must not be blindly merged.

8. EXPORTED STATE & POST-PROCESSING DIAGNOSTICS

The primary export to data/processed/stages_cleaned.xlsx was executed immediately following the company replacement sequence, outputting **2,292 rows × 23 columns**.

Subsequent notebook cells reloaded this file to execute further operations in memory:

- Column names were normalized to snake_case (accent stripping, lowercasing, non-alphanumeric conversion to underscores).
- The analytical column duree_calculee_mois was generated.
- Final verification confirmed 1,419 unique students across 2,292 records.

Data Persistence Note: No secondary export cell is recorded following the snake_case conversion or duration calculations. The processed Excel file on disk reflects the state prior to these final exploratory steps.

9. SUMMARY QUALITY STATUS & DOWNSTREAM RECOMMENDATIONS

Resolved Quality Items • Exact duplicates purged completely. • Invalid records (missing titles) dropped. • Primary company names standardized and deduplicated via hybrid fuzzy mapping. • Student ID, name, and convention consistency verified without errors.	Unresolved Limitations • Location fields remain >70% incomplete. • Manual edits lack independent source documentation. • Date/duration discrepancies logged as diagnostics only; no corrections applied. • Certain postal codes are marked as approximate.
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Guidelines for Downstream SQL / RAG Development

1. **Core Entities:** The dataset is fully qualified for structuring student, stage category, and primary company dimensions.
2. **Location Dimensions:** Geographic attributes must be modeled as highly sparse attributes or isolated in optional lookup tables due to missingness.
3. **Fuzzy Data Handling:** Company mappings should be treated as heuristic rather than ground-truth entity resolutions in vector search indexing.